



Universität
Zürich^{UZH}

Institute of Psychology



Bayesian hierarchical Signal Detection Theory in brms

Satellite Event: Analyzing data on the level of cognitive processes

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ESCoP 2025



Workshop: Overview

Monday

Tuesday

Morning

Intro
Why
hierarchical
models?

Signal-
Detection
Models
(brms)

Diffusion
Decision
Model
(bmm)

Practical
Exercises

Materials are
available via
GitHub

30 Min. Coffee
Breaks
10:30 & 15:00
60 Min. Lunch
Break
12:30

Afternoon

Memory
Measurement
Model
(bmm)

Practical
Exercises

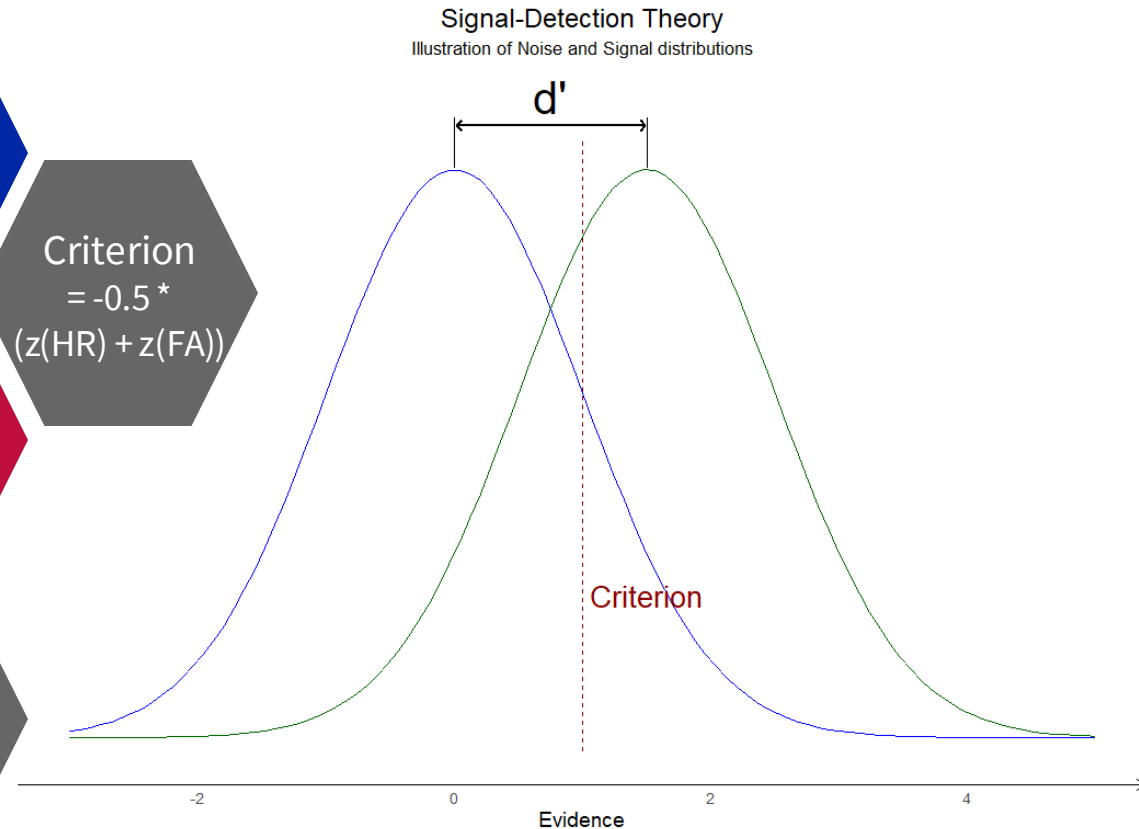
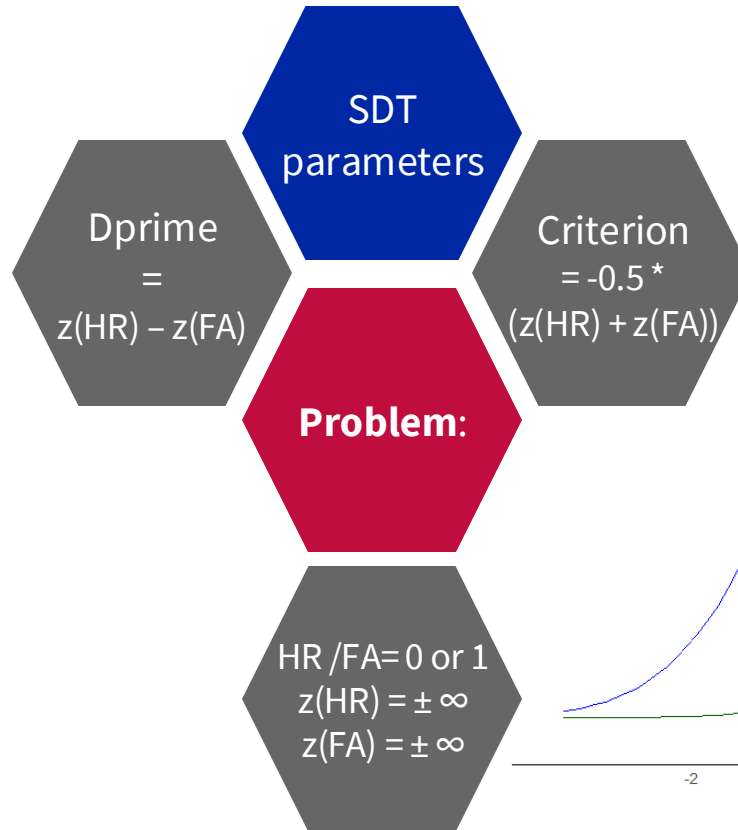
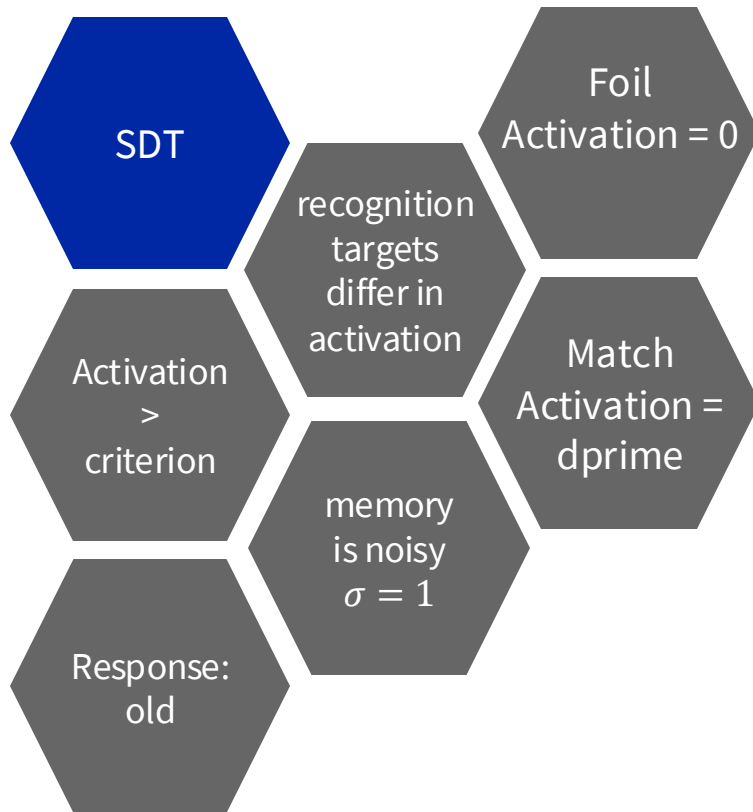
Signal
Discrimination
Model

Reporting
Results from
Cognitive
Modelling
Analyses

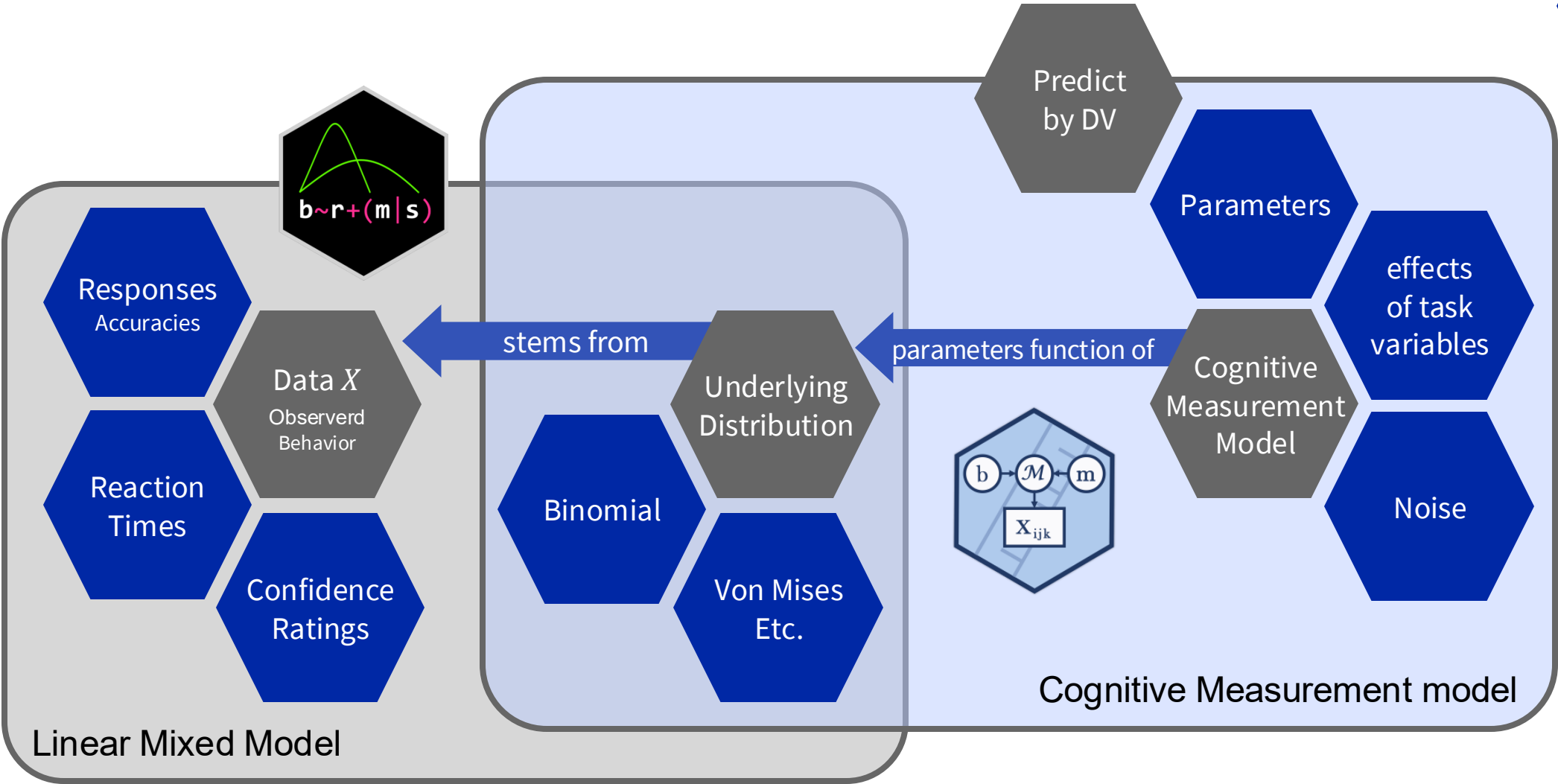
Ask questions
whenever
something is
not clear
😊

Repetition: Signal Detection Theory

Signal-
Detection
Models
(brms)



Data Analytic Model



Signal Detection Theory as a GLM

Signal-
Detection
Models
(brms)

SDT
= GLM

Data
Stems
From
Binomial

Noise Dist.
CDF =
Link
Function

Theta
=
 $f(dprime, \text{Criterion}, \text{stim_type})$

$$N_{sayold} \sim \text{Binom}(p_i, N_{trials})$$

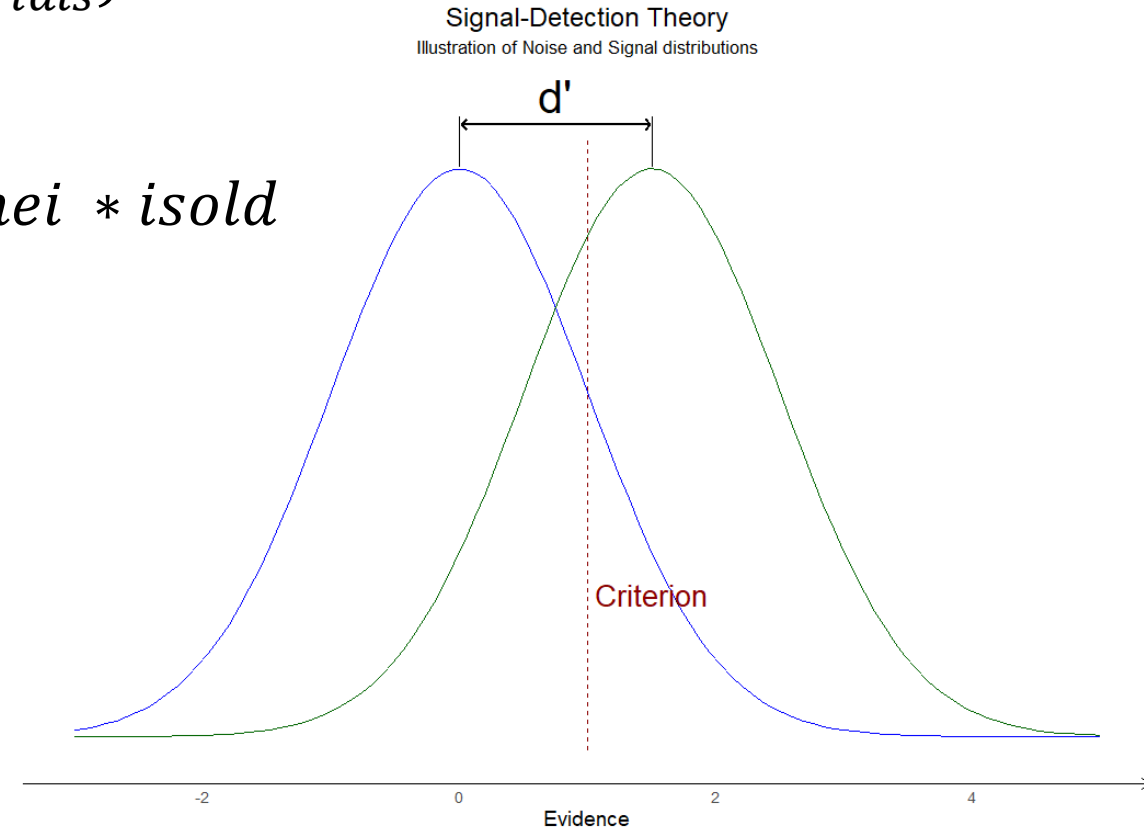
$$p_i = \Phi(\theta_i)$$

$$\theta_i = -\text{criterion}_i + dprime_i * \text{isold}$$

isold

Foil = 0

Target = 1



Signal Detection Theory as a GLM

Signal-
Detection
Models
(brms)

SDT
= GLM

Data
Stems
From
Binomial

Noise Dist.
=
Link
Function

Theta
=
f(dprime,
Criterion,
stim_type)

$$N_{sayold} \sim \text{Binom}(p_i, N_{trials})$$

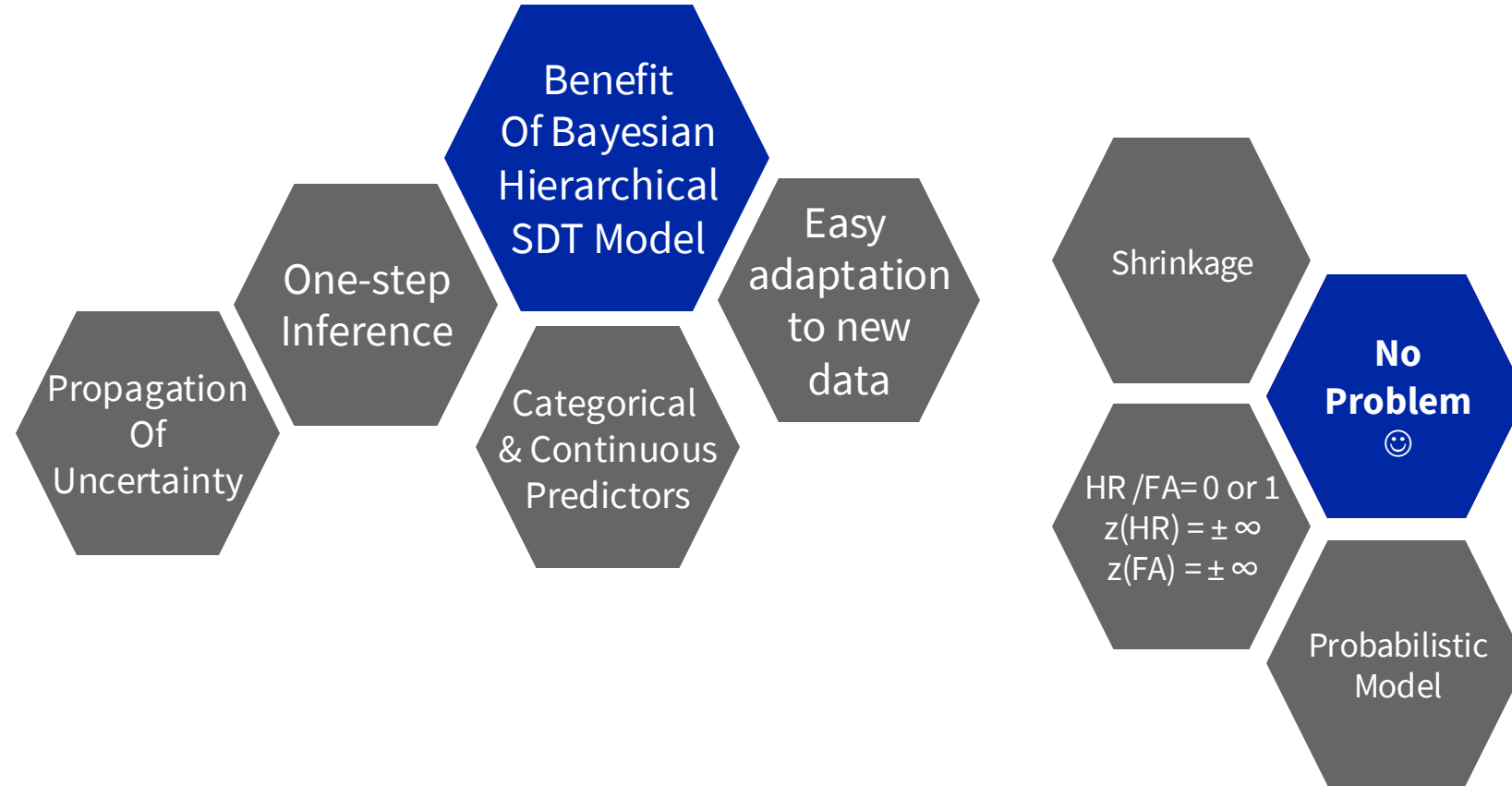
$$p_i = \Phi(\theta_i)$$

$$\theta_i = -\text{criterion}_i + dprime_i * isold$$

brms
formula

```
sdt_formula <- brmsformula(  
  n_sayold | trials(n_trials) ~ -crit + dprime*stim_isold,  
  dprime ~ 1 + predictors + (1 + predictors | id),  
  crit ~ 1 + predictors + (1 + predictors | id),  
  nl = TRUE  
)  
  
sdt_formula$family <- binomial(link = "probit")
```

Signal Detection Theory as a GLM



Signal-Detection Models (brms)



What are your questions so far?

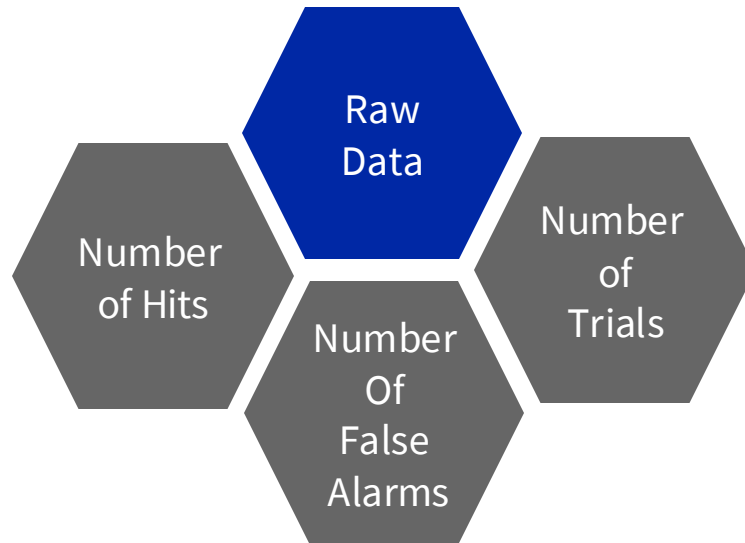


Signal Detection Theory in brms

Exercise II A



Signal-
Detection
Models
(brms)



Experimental Manipulation

Adaptive Instruction
→ Answer old & new equally

Exercise II B

1. Load the Williams et al. (2022) non aggregated data
2. Specify the SDT model in brms assuming a Binomial distribution with a probit link function
3. Fit the model, and try to conclude if there is an effect of the manipulation on d' and the criterion